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Introduction to Statistical Methods (ISM)

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Session 8
(13th December 2025)

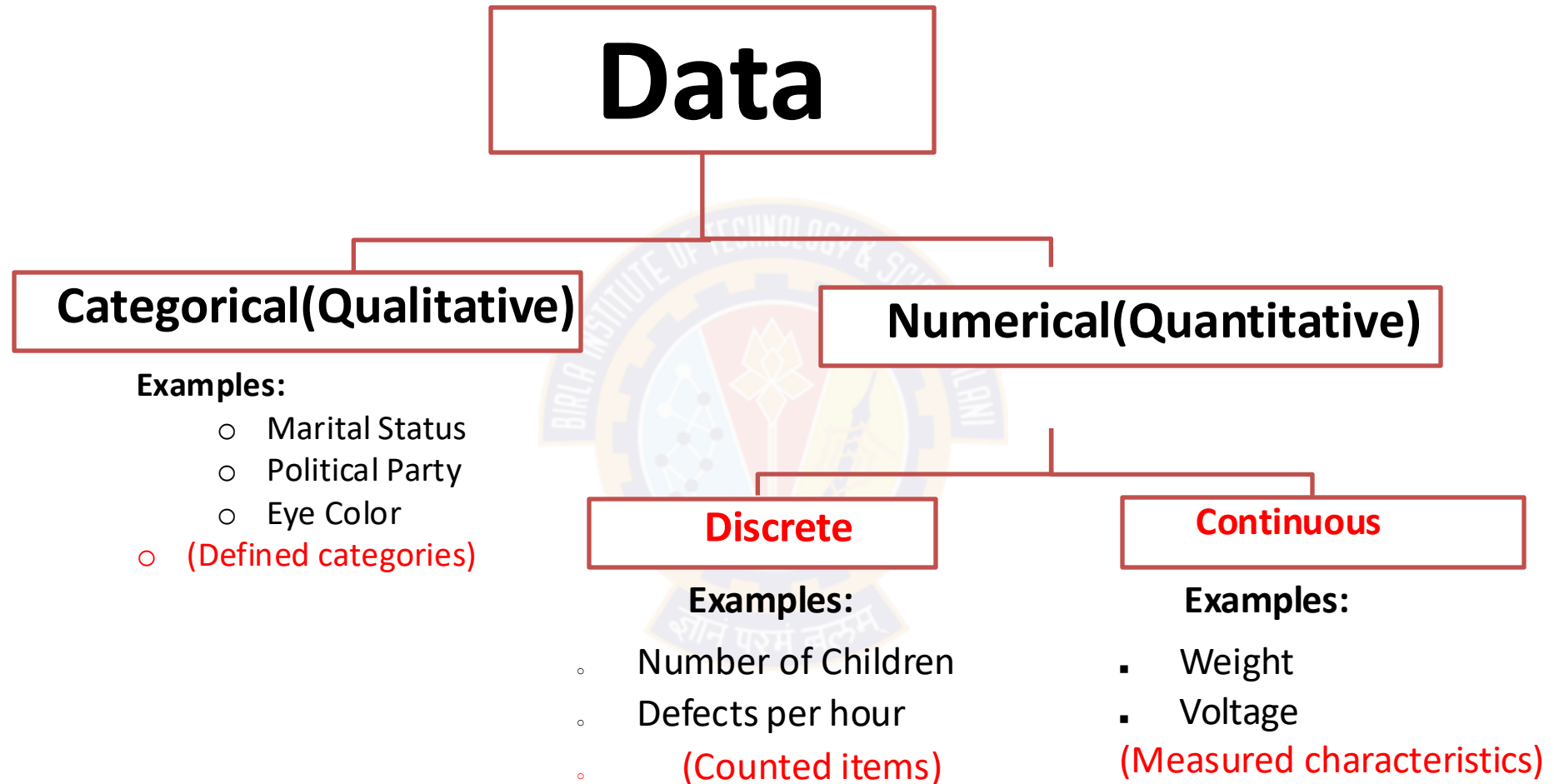


Formula Compendium & Review Questions

FORMULA COMPENDIUM



Data - Types



Variables

Types of Variable

Qualitative (Categorical): express a qualitative attribute such as hair color, eye color, religion.

Quantitative(Numerical): measured in terms of numbers such as height, weight, number of people.

Nominal:
No ordering is possible such as hair color, eye color, religion.

Ordinal:
Ordering is possible such as health, which can take values such as poor, reasonable, good, or excellent.

Discrete: countable and have a **finite number of possibilities** such as number of people

Continuous: not countable and have an **infinite number of possibilities** such as height

INTERVAL: ratio of values of variable do not have any meaning and it does not have an inherently defined zero value such as temperature

RATIO: ratio of values of variable have meaning and it have an inherently defined zero value such as length

Measures of Central Tendency

Measure	Formula
Mean	$\mu = \frac{\sum_{i=1}^n x_i}{n}$
Median	$p = \frac{n+1}{2}$
Mode	The mode is the value or values that occur most frequently in the data set. A data set can have more than one mode, and it can also have no mode

A unimodal mode is a set of data with only one mode.

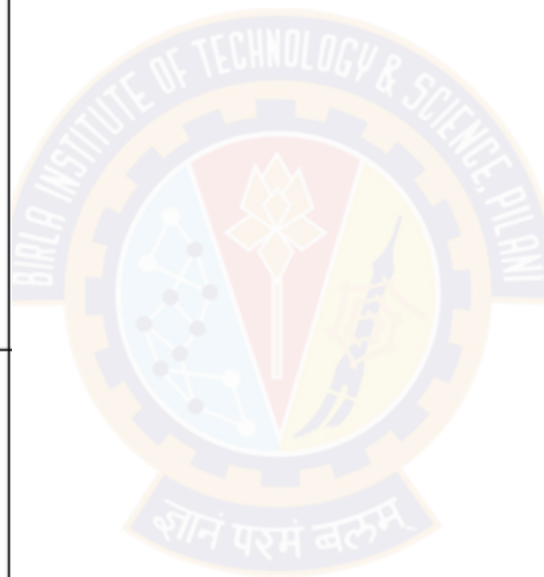
A bimodal mode is a set of data that has two modes.

A trimodal mode is a set of data that has three modes.

A given set of data may have one or more than one Mode. A set of numbers with one Mode is unimodal, a set of numbers having two Modes is bimodal, a set of numbers having three Modes is trimodal, and any set of numbers having four or more than four Modes is known as multimodal.

Measures of Variability

Range	Range = $x_n - x_1$
Variance	For a Population $\sigma^2 = \frac{\sum_{i=1}^n (x_i - \mu)^2}{n}$ For a Sample $s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$
Standard deviation	For a Population $\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \mu)^2}{n}}$ For a Sample $s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$



$$\begin{aligned}
 \sigma^2 &= \frac{\sum (X - \mu)^2}{N} \\
 &= \frac{\sum (X^2 - 2\mu X + \mu^2)}{N} \\
 &= \frac{\sum X^2}{N} - \frac{2\mu \sum X}{N} + \frac{N\mu^2}{N} \\
 &= \frac{\sum X^2}{N} - 2\mu^2 + \mu^2 \\
 &= \frac{\sum X^2}{N} - \mu^2
 \end{aligned}$$

Data distribution-Skew (Asymmetry) & Symmetry

➤ Skewed

○ Negatively (Left): $\text{mean} < \text{median} < \text{Mode}$

- Distributions with a left skew have long left tails;

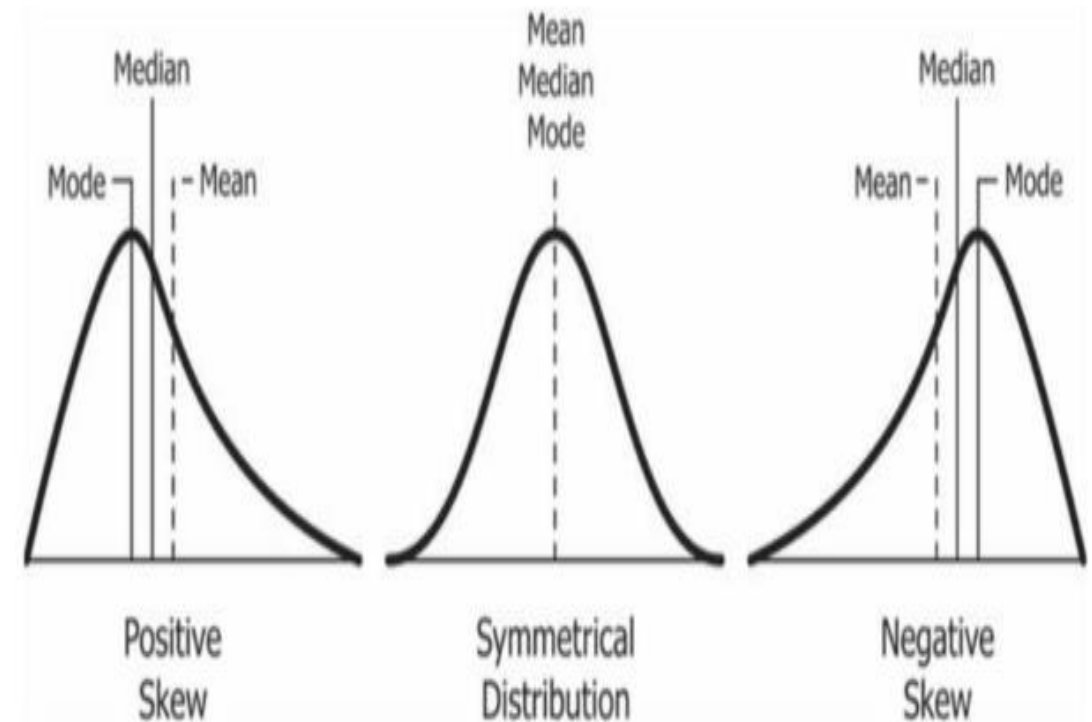
○ Positively (Right) : $\text{mean} > \text{median} > \text{Mode}$

- Distributions with a right skew have long right tails.

➤ Symmetry: $\text{Mean} = \text{Median} = \text{Mode}$

➤ Bimodal : has two distinct modes

➤ Multi-modal : has more than 2 distinct modes

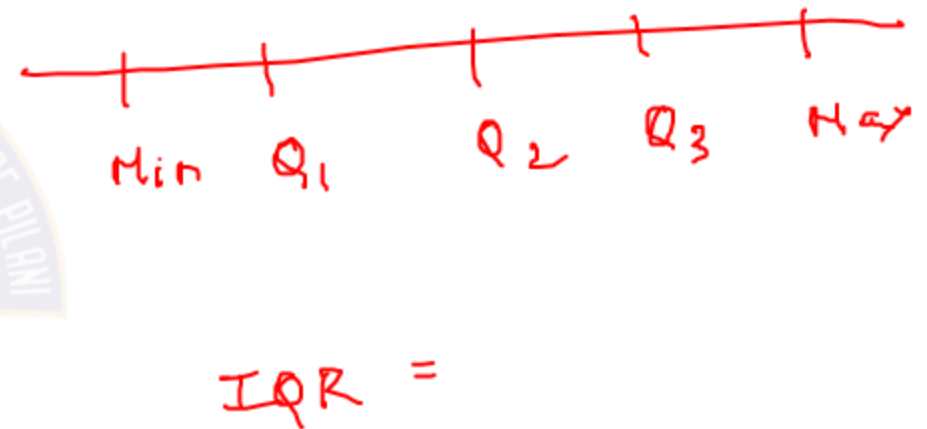


Note: In science, an empirical relationship is a relationship that is supported by experiment and observation but not necessarily supported by theory. For moderately skewed/unskew data, an empirical relationship is observed as $\text{Mode} = 3\text{Median} - 2\text{Mean}$. Further Mean is applied for unskew/moderately skewed data whereas median is a preferred measure for heavily skewed data.

Five-point summary of Data

The five number summary of data includes 5 items:

- **Minimum.**
- **Q1** (the first quartile, or the 25% mark).
- **Median.** = Q_2 (50%)
- **Q3** (the third quartile, or the 75% mark).
- **Maximum.**



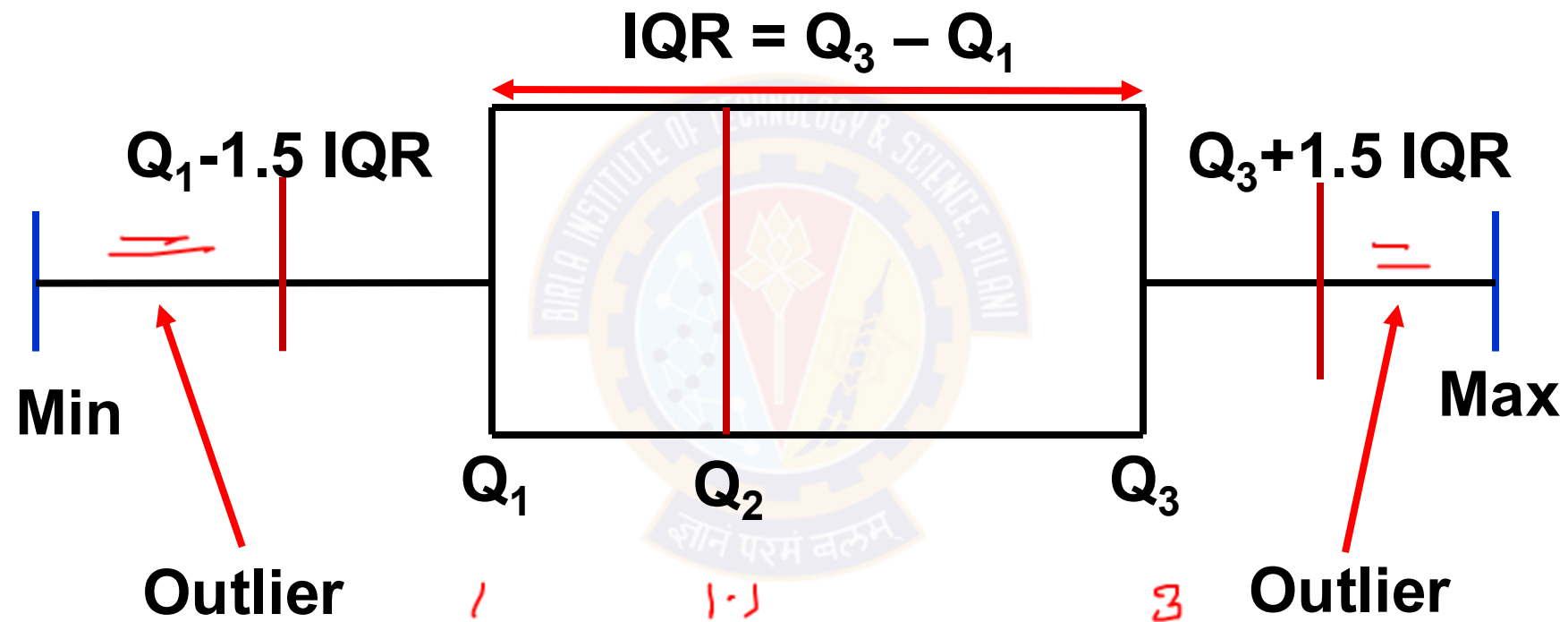
Interquartile range (IQR) & Quartile Deviation (QD) & Outliers

- It is measure of Variation
- Also Known as Mid-spread : Spread in the Middle 50%
- Difference Between Third & First Quartiles:
- Not Affected by Extreme Values
- Interquartile Range = $IQR = Q_3 - Q_1$
- Quartile Deviation = $IQR/2 = (Q_3 - Q_1)/2$

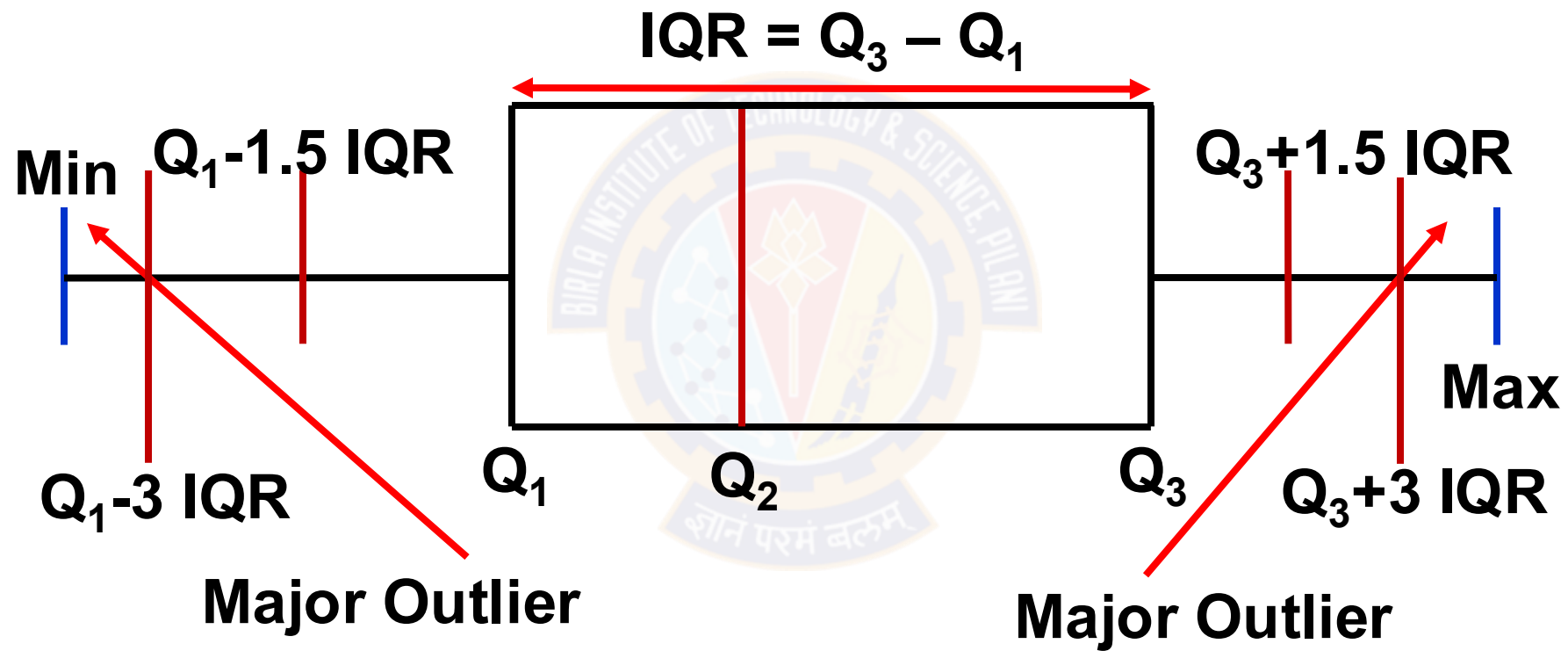


Quartiles	Quartiles separate a data set into four sections. The median is the second quartile Q_2 . It divides the ordered data set into higher and lower halves. The first quartile, Q_1 , is the median of the lower half not including Q_2 . The third quartile, Q_3 , is the median of the higher half not including Q_2 .
Interquartile range	$IQR = Q_3 - Q_1$
Potential outliers, if any.	$\text{Upper Fence} = Q_3 + 1.5 \times IQR$ $\text{Lower Fence} = Q_1 - 1.5 \times IQR$

Box and Whisker plot



Box and Whisker plot



Definition of Probability

- **Classical approach:**

CLASSICAL PROBABILITY

$$\text{Probability of an event} = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

Axiomatic approach

Probability is a number that is assigned to each member of a collection of events from a random experiment that satisfies the following properties:

If S is the sample space and E is any event in a random experiment,

(1) $P(S) = 1$

(2) $0 \leq P(E) \leq 1$

(3) For two events E_1 and E_2 with $E_1 \cap E_2 = \emptyset$

$$P(E_1 \cup E_2) = P(E_1) + P(E_2)$$

The Addition Rule

- The probability that event A or B will occur is given by

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

“if” $A \cap B = \emptyset$

- If events A and B are **mutually exclusive**, then the rule can be simplified to

$$P(A \cup B) = P(A) + P(B).$$

Counting Rules, Probability and Venn Diagram

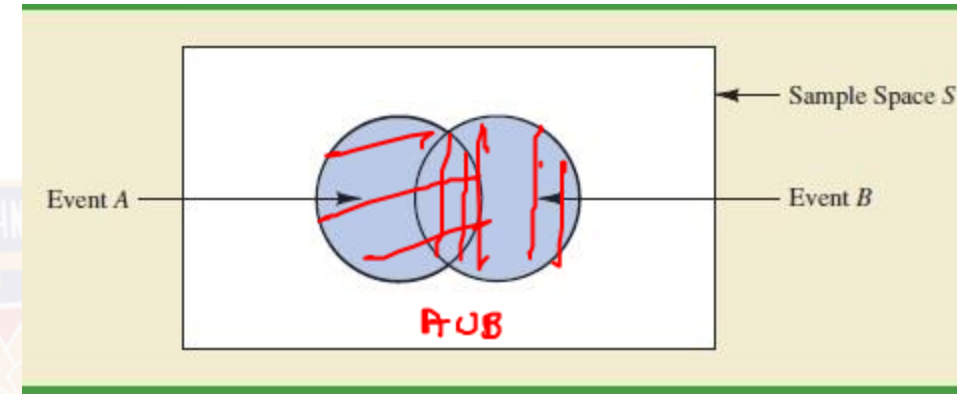
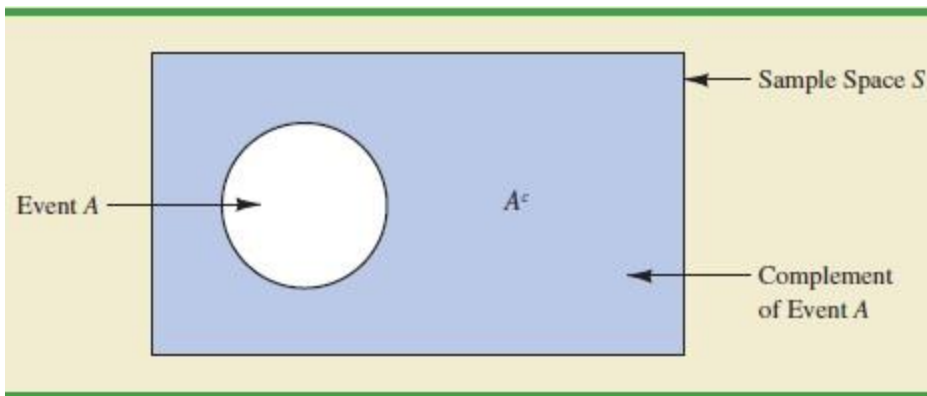
Counting Rule for Combinations

$$C_n^N = \binom{N}{n} = \frac{N!}{n!(N-n)!}$$

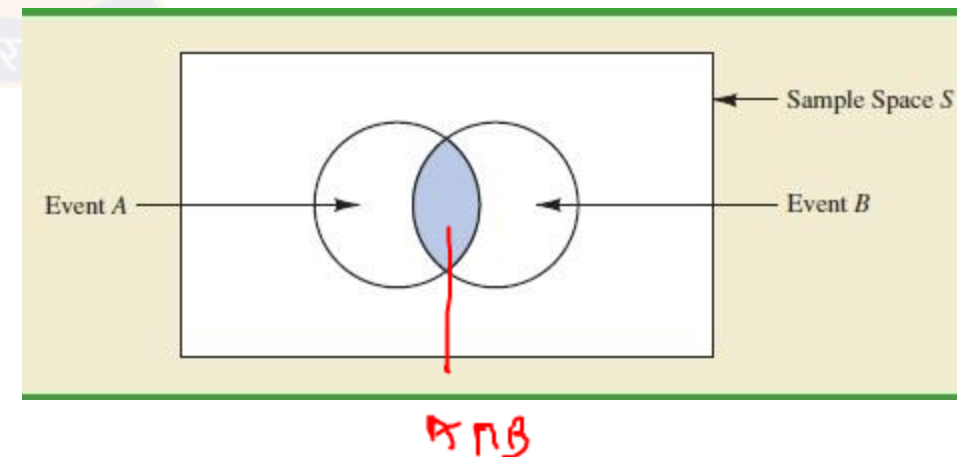
Counting Rule for Permutations

$$P_n^N = n! \binom{N}{n} = \frac{N!}{(N-n)!}$$

$$P(A) = 1 - P(A^c)$$



$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



Independent Events / ~~Mutually exclusive events~~

A and B are independent if and only if (iff)

$$P(A \cap B) = P(A) \cdot P(B)$$



Conditional Probability

- Let A and B be two events in a sample space. The conditional probability that event B occurs given that event A has occurred is denoted by B / A (B given A – not to read this as B by A) and its probability is

$$P(B|A) = \frac{P(A \text{ and } B)}{P(A)}$$



- The conditional probability that event A occurs given that event B has occurred is denoted by A / B and its probability is

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$= P(A) \Rightarrow A \& B$
 $\swarrow \searrow$ are indep
 $P(A \cap B) = P(A) \cdot P(B)$

- From these we can also have

$$\begin{aligned} P(A \cap B) &= P(A) P(B|A) & P(A) \neq 0 \\ &= P(B) P(A|B) & P(B) \neq 0 \end{aligned}$$

Multiplication rule

- Let A, B and C be three events in a sample space S, then $P(A \cap B \cap C) = P(A) P(B|A) P(C|A \cap B)$ and it is called multiplication rule

In general, $A_1, A_2, \dots, A_n$ are events in S then

$$P(A_1 \cap A_2 \cap \dots \cap A_n)$$

$$= P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \dots P(A_n|A_1 \cap A_2 \cap \dots \cap A_{n-1})$$

Independent Events

An important result from the conditional probability:

If B has no effect on A, then, $P(A/B) = P(A)$

Also $P(B/A) = P(B)$

and we say the events are independent. (Statistically Independent)
i.e., The probability of A does not depend on B.

so,
$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

becomes,
$$P(A) = \frac{P(A \cap B)}{P(B)}$$

or
$$P(A \cap B) = P(A) \times P(B)$$

Total Probability

The Law of Total Probability:

Let $A_1, A_2, \dots, A_k$ be mutually exclusive and exhaustive events.

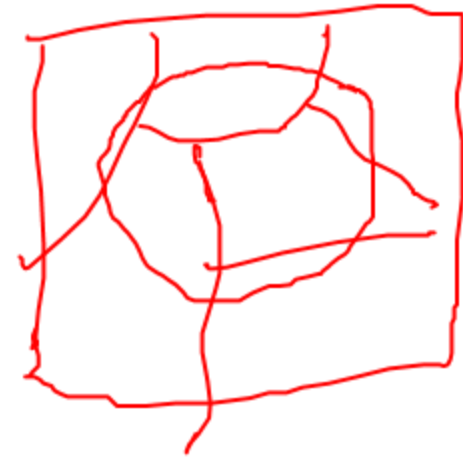
Then for any other event B ,

disjoint

Union = Ω = S

$$P(B) = P(B|A_1)P(A_1) + P(B|A_2)P(A_2) \dots + P(B|A_k)P(A_k)$$

$$P(B) = \sum_{i=1}^k P(B|A_i)P(A_i)$$



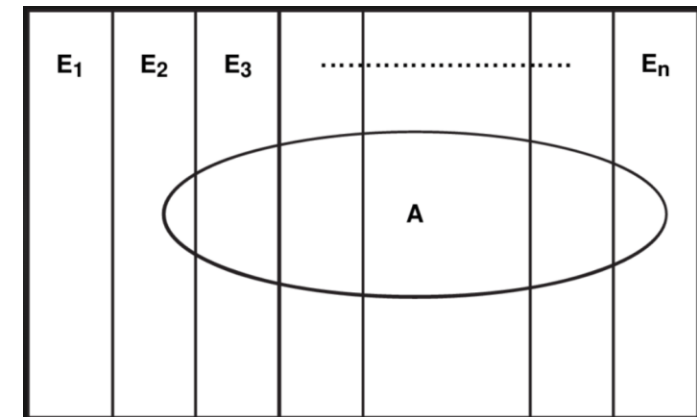
BAYE'S THEOREM:

Suppose that $E_1, E_2, \dots, E_n$ are mutually exclusive events of a sample space "S"

such that $P(E_i) > 0$ for $i = 1, 2, 3, \dots, n$ and A is any arbitrary event of "S"

such that $P(A) > 0$ and $A \subseteq \bigcup_{i=1}^n E_i$ then the conditional probability of E_i given A

for $i = 1, 2, 3, \dots, n$ is given by $P(E_i/A) = \frac{P(E_i)P(A/E_i)}{\sum_{i=1}^n P(E_i)P(A/E_i)}$



Naïve Bayes for Text Classification / Laplace Smoothing

- Naïve Bayes is commonly used for **text classification**
- For a document with k terms $d = (t_1, \dots, t_k)$

Fraction of documents in c

$$P(c|d) = P(c)P(d|c) = P(c) \prod_{t_i \in d} P(t_i|c)$$

- $P(t_i|c)$ = Fraction of terms from **all documents** in c that are t_i .

Number of times t_i appears in some document in c

$$P(t_i|c) = \frac{N_{ic} + 1}{N_c + T}$$

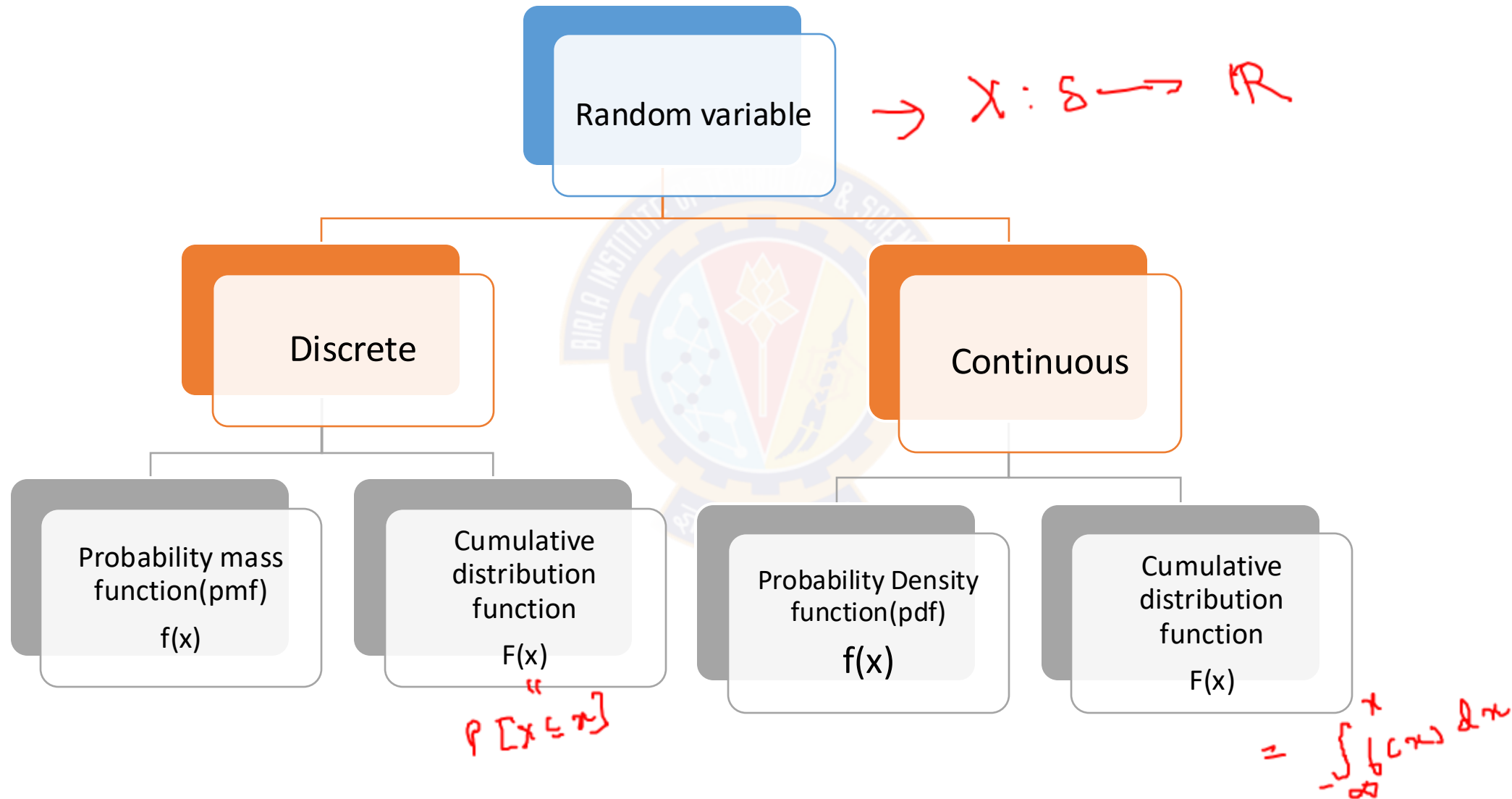
Laplace Smoothing

Total number of terms in all documents in c

Number of unique words (vocabulary size)

- Easy to implement and works relatively well
- **Limitation**: Hard to incorporate **additional features** (beyond words).
 - E.g., number of adjectives used.

Classification of Random Variables



Discrete Probability Distributions

- The probability distribution for a random variable describes how probabilities are distributed over the values of the random variable.
- The probability distribution is defined by a probability function, denoted by $f(x)$ or $p(x)$, which provides the probability for each value of the random variable.
- The required conditions for a discrete probability function are:

$$f(x) \geq 0$$

$$\sum f(x) = 1$$

- We can describe a discrete probability distribution with a **table, graph, or equation**.
- Advantage: once the probability distribution is known, it is relatively easy to determine the probability of a variety of events that may be of interest to the decision maker.

Discrete Uniform Probability Distribution

➤ The discrete uniform **probability** distribution is the simplest example of a discrete probability distribution given by a formula.

○ The discrete uniform probability function is

$$f(x) = 1/n$$

where:

n = the number of values the random variable may assume

- Note that the values of the random variable are equally likely.
- Example: A discrete RV X : No. on a dice has the uniform probability

$$f(x) = 1/6 \text{ for } x = 1, 2, 3, 4, 5, 6$$

Mathematical Expectation

- The mean or Expected value of a discrete random variable:

$$E(X) = \mu = \sum x f(x)$$

- Multiplying RV by a constant a , $E(aX) = aE(X)$
- Adding a constant b , $E(X + b) = E(X) + b$
- Therefore, $E(aX + b) = ?$
- $E(X/Y) \neq \frac{E(X)}{E(Y)}$
- $E(X.Y) \neq E(X).E(Y)$ unless they are independent

Variability of Discrete Random Variables

➤ The **variance** of a **discrete random variable** x is

$$\text{var}(X) = \sigma^2 = E(\underbrace{X - \mu})^2 = \sum (x - \mu)^2 f(x)$$

➤ Or

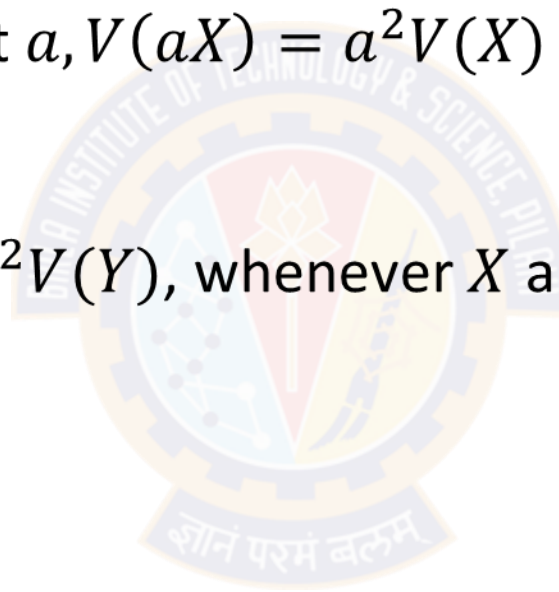
$$\begin{aligned}\text{var}(X) = \sigma^2 &= E(X^2) - (E(X))^2 \\ &= E(X^2) - \mu^2 \\ &= \sum x^2 f(x) - \mu^2\end{aligned}$$

➤ The **standard deviation** (σ), is defined as the positive square root of the variance.

$$\begin{aligned}E(X) &= \sum x f(x) \\ &\text{or } \int x f(x) dx \\ E(h(x)) &= \sum h(x) f(x) \\ &\text{or } \int h(x) f(x) dx\end{aligned}$$

Rules of variability

- $V(b) = 0$, b is the constant
- Multiplying RV by a constant a , $V(aX) = a^2V(X)$
- Therefore, $V(aX+b) = ?$
- $V(aX + bY) = a^2V(X) + b^2V(Y)$, whenever X and Y are independent
- $\sigma_{aX} = |a|\sigma_x$



Continuous Random Variables

Probability Density Function

For a continuous random variable X , a **probability density function** is a function such that

$$(1) \quad f(x) \geq 0$$

$$(2) \quad \int_{-\infty}^{\infty} f(x) dx = 1$$

$$(3) \quad P(a \leq X \leq b) = \int_a^b f(x) dx = \text{area under } f(x) \text{ from } a \text{ to } b \text{ for any } a \text{ and } b \quad (4.1)$$

Continuous Random Variables

Mean and Variance

Suppose that X is a continuous random variable with probability density function $f(x)$. The **mean** or **expected value** of X , denoted as μ or $E(X)$, is

$$\mu = E(X) = \int_{-\infty}^{\infty} xf(x)dx$$

The **variance** of X , denoted as $V(X)$ or σ^2 , is

$$\sigma^2 = V(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)dx = \int_{-\infty}^{\infty} x^2 f(x)dx - \mu^2$$

The **standard deviation** of X is $\sigma = \sqrt{\sigma^2}$.

Joint Probability Distribution Function

- ❖ Let $X = \{x_1, x_2, \dots, x_m\}$ and $Y = \{y_1, y_2, \dots, y_n\}$ be two discrete random variables. Then $P(x, y) = J_{ij}$ is called joint probability function of X and Y if it satisfies the conditions:

$$(i) J_{ij} \geq 0 \quad (ii) \sum_{i=1}^m \sum_{j=1}^n J_{ij} = 1$$

- ❖ Set of values of this joint probability function J_{ij} is called joint probability distribution of X and Y.

	y_1	y_2	...	y_n	Sum
x_1	J_{11}	J_{12}	...	J_{1n}	$f(x_1)$
x_2	J_{21}	J_{22}	...	J_{2n}	$f(x_2)$
...	...	...	...	...	...
x_m	J_{m1}	J_{m2}	...	J_{mn}	$f(x_m)$
Sum	$g(y_1)$	$g(y_2)$	...	$g(y_n)$	Total = 1

Joint Density Function

When X and Y are continuous random variables, the **joint density function** $f(x, y)$ is a surface lying above the xy plane, and $P[(X, Y) \in A]$, where A is any region in the xy plane, is equal to the volume of the right cylinder bounded by the base A and the surface.

The function $f(x, y)$ is a **joint density function** of the continuous random variables X and Y if

1. $f(x, y) \geq 0$, for all (x, y) ,
2. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \, dx \, dy = 1$,
3. $P[(X, Y) \in A] = \int \int_A f(x, y) \, dx \, dy$, for any region A in the xy plane.

Marginal Distributions

The **marginal distributions** of X alone and of Y alone are

$$g(x) = \sum_y f(x, y) \quad \text{and} \quad h(y) = \sum_x f(x, y)$$

for the discrete case, and

$$g(x) = \int_{-\infty}^{\infty} f(x, y) \, dy \quad \text{and} \quad h(y) = \int_{-\infty}^{\infty} f(x, y) \, dx$$

for the continuous case.



Independent Random Variables

- Let X and Y are two random variables with joint probability function $f(x, y)$ are said to be independent if following condition satisfied:
- $f(x, y) = g(x).h(y)$ for all x and y , where $g(x)$ is marginal probability function of X and $h(y)$ is marginal probability function of Y .

Example: Suppose $f(x, y) = e^{-(x+y)}$, $x \geq 0, y \geq 0$. For this probability function, marginal probability function for X is $g(x) = e^{-x}$, $x \geq 0$ and marginal probability function for Y is $h(y) = e^{-y}$, $y \geq 0$.

Clearly $f(X, Y) = g(X).h(Y)$. So X and Y are independent variables.

Conditional Probability Distribution Function

- Discrete Case:

$$P[X = x_i / Y = y_j] = \frac{P[X = x_i \text{ and } Y = y_j]}{P[Y = y_j]} = \frac{p_{ij}}{p_{.j}}$$

A B B
?
p_{ij}

The set $\{x_i, p_{ij} / p_{.j}\}$, $i = 1, 2, 3, \dots$ is called the conditional probability distribution of X given $Y = y_j$.

The conditional probability function of Y given $X = x_i$ is given by

$$P[Y = y_j / X = x_i] = \frac{P[Y = y_j \text{ and } X = x_i]}{P[X = x_i]} = \frac{p_{ij}}{p_{i.}}$$

The set $\{y_j, p_{ij} / p_{i.}\}$, $j = 1, 2, 3, \dots$ is called the conditional probability distribution of Y given $X = x_i$.

Conditional Probability Density Function

$$\underline{f(y \mid x)} = \frac{f(x, y)}{f_X(x)} \quad \text{or} \quad f(x \mid y) = \frac{f(x, y)}{f_Y(y)}$$

- the continuous version of Bayes' theorem

$$f(y \mid x) = \frac{f(x \mid y)f_Y(y)}{f_X(x)}$$

- another expression of the marginal pdf

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y)dy = \int_{-\infty}^{\infty} f(x \mid y)f_Y(y)dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y)dx = \int_{-\infty}^{\infty} f(y \mid x)f_X(x)dx$$

Bernoulli Distribution

Definition

A random variable 'X' is said to have Bernoulli distribution if its probability mass function is given by

$$p(x) = \begin{cases} p^x q^{1-x}, & x = 0, 1 \\ 0, & \text{elsewhere} \end{cases}$$

This represents a single trial with two possible outcomes (usually labeled as success/failure or 1/0) where success has a probability p and failure has a probability $q = 1 - p$.

Binomial Distribution

Definition

A random variable 'X' is said to have Binomial distribution if its probability mass function is given by

$$p(x) = \begin{cases} \binom{n}{x} p^x q^{n-x}, & x = 0, 1, 2, 3, \dots, n \\ 0, & \text{elsewhere} \end{cases}$$

It describes the number of successes in a fixed number of independent Bernoulli trials. It involves parameters n (number of trials) and p (probability of success).

Poisson Distribution

➤ Poisson distribution is suitable for **rare events** for which the probability of occurrence 'p' is very small and the number of trials 'n' is very large.

➤ Also, binomial distribution can be approximated by Poisson distribution when $n \rightarrow \infty$ and $p \rightarrow 0$ such that $\lambda = np = \text{constant}$

The approximation is more accurate if $n \geq 100$ and as long as $np \leq 10$

Poisson Distribution

Definition

A random variable 'X' is said to have Poisson distribution if its probability mass function is given by

$$p(x) = \begin{cases} e^{-\lambda} \frac{\lambda^x}{x!}, & x = 0, 1, 2, \dots \\ 0, & \text{elsewhere} \end{cases}$$

- Poisson distribution is the discrete probability distribution of a discrete random variable X, which may not have upper bound. It is defined for non negative values of x .

Poisson approximation to Binomial distribution :

When n is large and p is small, Binomial probabilities can be approximated by Poisson's distribution with mean $\lambda=np$.

The approximation is more accurate if $n \geq 100$ and as long as $np \leq 10$



Normal Distribution

A continuous random variable X which assumes all possible values in the entire real space, i.e., $-\infty < x < \infty$ is said to follow normal distribution with two parameters such as mean μ and variance σ^2 if it has the probability density function given by

$$f(x; \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

μ = mean

σ = standard deviation

π = 3.14159

e = 2.71828

$$-\infty < x < \infty$$
$$-\infty < \mu < \infty, \sigma > 0$$

$$X \sim N(\mu, \sigma^2)$$
$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

$$P[a \leq x \leq b] = \int_a^b f(x) dx$$
$$P\left[\frac{a-\mu}{\sigma} \leq Z \leq \frac{b-\mu}{\sigma}\right] = F\left(\frac{b-\mu}{\sigma}\right) - F\left(\frac{a-\mu}{\sigma}\right)$$

Tabulate $F(z)$

Normal Distribution: Mean and Variance

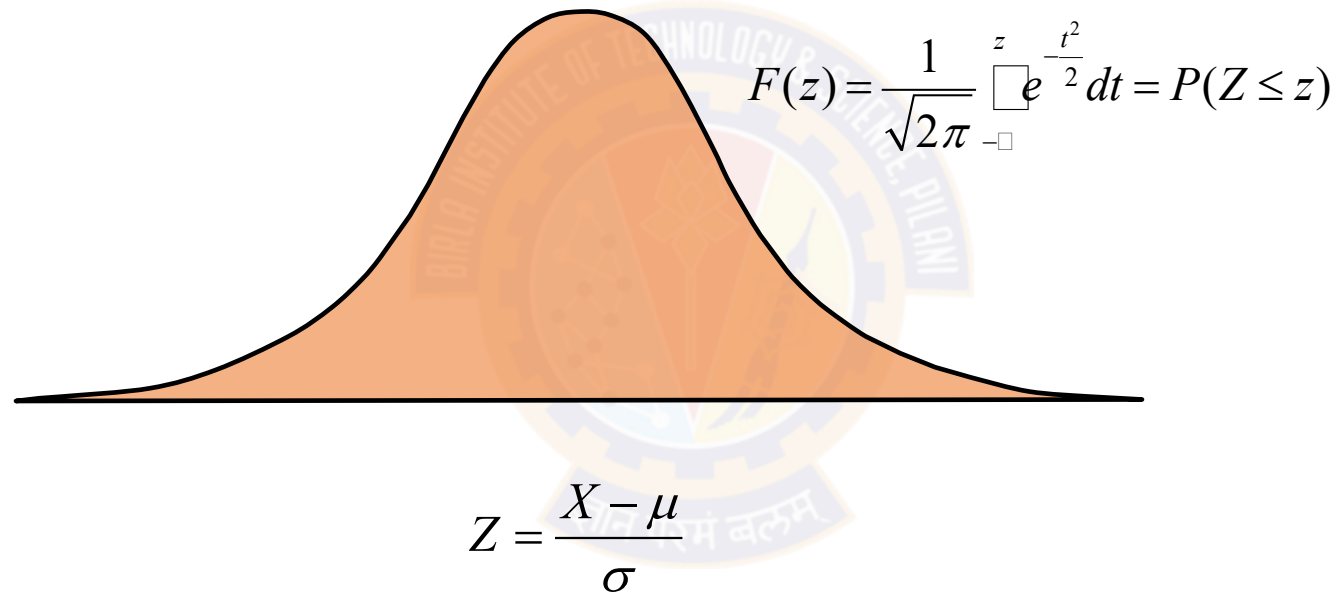
$$E(X)=\mu = \int_{-\infty}^{+\infty} x \frac{1}{\sigma \sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$\text{Var}(X)=\sigma^2 = \left(\int_{-\infty}^{+\infty} x^2 \frac{1}{\sigma \sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \right) - \mu^2$$

Standard Deviation(X)= σ

Standard Normal Distribution

The **standard normal distribution** is a normal distribution with a mean of 0 and a standard deviation of 1.



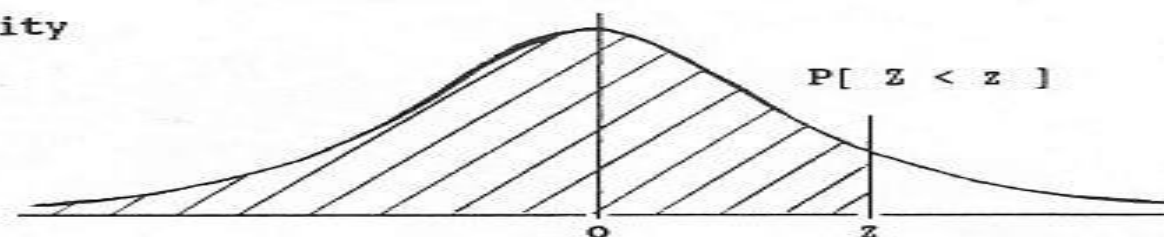
All normal distributions can be converted into the standard normal curve by subtracting the mean and dividing by the standard deviation.

STANDARD STATISTICAL TABLES

1. Areas under the Normal Distribution

The table gives the cumulative probability up to the standardised normal value z i.e.

$$P[Z < z] = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} \exp(-\frac{1}{2}z^2) dz$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5159	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7854
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8804	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9773	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9865	0.9868	0.9871	0.9874	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9924	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9980	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
z	3.00	3.10	3.20	3.30	3.40	3.50	3.60	3.70	3.80	3.90
P	0.9986	0.9990	0.9993	0.9995	0.9997	0.9998	0.9998	0.9999	0.9999	1.0000

Normal Approximation to a Binomial Distribution

The normal distribution is used to approximate the binomial distribution **when it would be impractical to use the binomial distribution** to find a probability.

Normal Approximation to a Binomial Distribution

If $np \geq 15$ and $nq \geq 15$, then the binomial random variable x is approximately normally distributed with mean

and standard deviation
 $\mu = np$

$$\sigma = \sqrt{npq}.$$

REVIEW QUESTIONS



Question 1

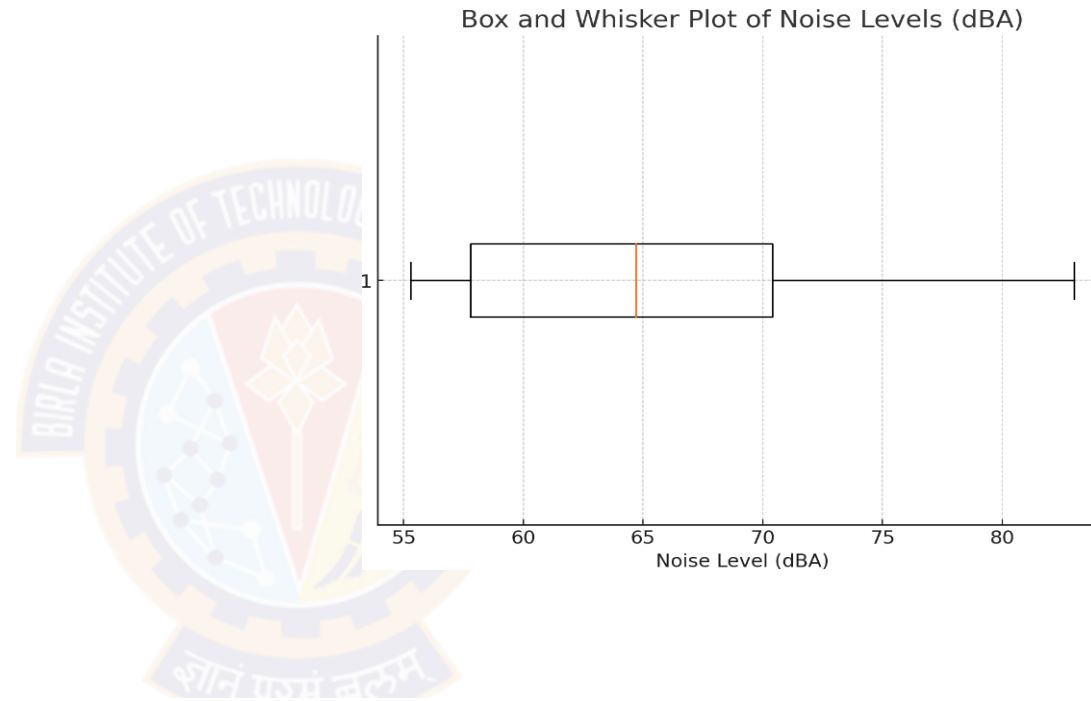
- **Q1.** A sample of 77 individuals working at a particular office was selected and the noise level (dBA) experienced by everyone is the following data:

55.3, 55.3, 55.3, 55.9, 55.9, 55.9, 55.9, 56.1, 56.1, 56.1, 56.1, 56.1, 56.1,
56.8, 56.8, 57.0, 57.0, 57.0, 57.8, 57.8, 57.8, 57.9, 57.9, 57.9, 58.8, 58.8,
58.8, 59.8, 59.8, 59.8, 62.2, 62.2, 63.8, 63.8, 63.8, 63.9, 63.9, 63.9, 64.7,
64.7, 64.7, 65.1, 65.1, 65.1, 65.3, 65.3, 65.3, 65.3, 67.4, 67.4, 67.4, 67.4,
68.7, 68.7, 68.7, 68.7, 69.0, 70.4, 70.4, 71.2, 71.2, 71.2, 73.0, 73.0, 73.1,
73.1, 74.6, 74.6, 74.6, 74.6, 79.3, 79.3, 79.3, 79.3, 83.0, 83.0, 83.0.

- Find a) Arithmetic Mean, SD, variance, and IQR
b) Draw box and whisker plot
c) Comment on the outliers, if any.

Solution:

- Arithmetic Mean: 64.89 dBA
- Standard Deviation (SD): 7.80 dBA
- Variance: 60.88 (dBA)^2
- Interquartile Range (IQR): 12.60 dBA



- Lower Bound: $Q1 - 1.5 \times IQR = 59.8 - (1.5 \times 12.6) = 40.9$
- Upper Bound: $Q3 + 1.5 \times IQR = 72.4 + (1.5 \times 12.6) = 91.3$

No data points fall below 40.9, but values such as 79.3 and 83.0 are potential upper outliers, as they exceed 91.3.

Question 2

Q2. In a factory, machine **A** produces **40%** of the output and machine **B** produces **60%**. On average, **9 items in 1000** produced by **A** are defective, and **1 item in 250** produced by **B** is defective. An item is drawn at random from a day's output and is found to be defective.

What is the probability that it was produced by A or by B?



Solution:

Output produced by A = 40%

$$P(A) = 0.4$$

Output produced by B = 60%

$$P(B) = 0.6$$

Probability that items produced by A are defective:

$$P(D | A) = \frac{9}{1000} = 0.009$$

Probability that items produced by B are defective:

$$P(D | B) = \frac{1}{250} = 0.004$$

Solution:

Using Bayes' Theorem:

Probability that the defective item was produced by A:

$$\begin{aligned}P(A | D) &= \frac{P(A) P(D | A)}{P(A) P(D | A) + P(B) P(D | B)} \\&= \frac{0.4 \times 0.009}{0.4 \times 0.009 + 0.6 \times 0.004} \\&= \frac{0.0036}{0.0036 + 0.0024} \\&= \frac{0.0036}{0.006} = 0.6\end{aligned}$$

Probability that the defective item was produced by B:

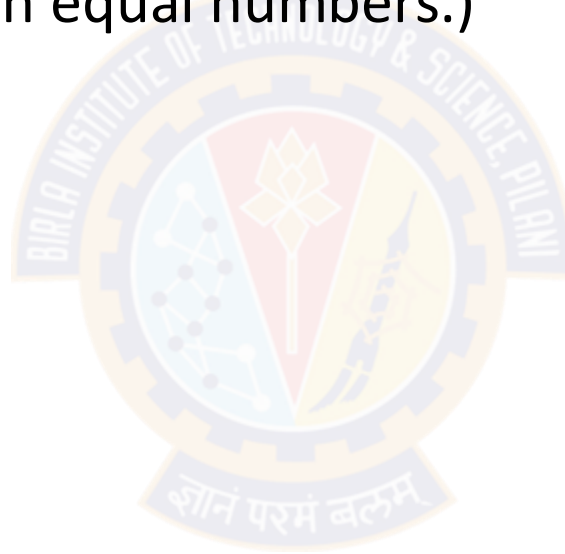
$$\begin{aligned}P(B | D) &= \frac{P(B) P(D | B)}{P(A) P(D | A) + P(B) P(D | B)} \\&= \frac{0.6 \times 0.004}{0.006} \\&= \frac{0.0024}{0.006} = 0.4\end{aligned}$$

Final Answer:

- Probability it was produced by **A = 0.6**
- Probability it was produced by **B = 0.4**

Question 3

Suppose **5 men out of 100** and **25 women out of 10,000** are colour blind. A colour-blind person is chosen at random. What is the probability that the person is **male**? (Assume males and females are in equal numbers.)



Solution:

Since males and females are equal in number:

$$P(M) = \frac{1}{2}, P(W) = \frac{1}{2}$$

Probability that a man is colour blind:

$$P(B | M) = \frac{5}{100} = 0.05$$

Probability that a woman is colour blind:

$$P(B | W) = \frac{25}{10000} = \frac{1}{400} = 0.0025$$

We want the probability that the chosen blind person is **male**, that is $P(M | B)$.

Using **Bayes' Theorem**:

$$P(M | B) = \frac{P(B | M) P(M)}{P(B | M) P(M) + P(B | W) P(W)}$$

Substitute the values:

$$P(M | B) = \frac{0.05 \times 0.5}{(0.05 \times 0.5) + (0.0025 \times 0.5)} = \frac{0.025}{0.025 + 0.00125} = \frac{0.025}{0.02625} = 0.95$$

Final Answer:

The probability that the chosen colour-blind person is **male** = **0.95** (95%).

Question 4:

Suppose we got the new message with the words '**Dear Friend**', Decide whether this new message is a normal or spam message?

i.e. Normal/ Dear, Friend = Yes or No

Note if we know $P(\text{Normal/ Dear, Friend})$ and $P(\text{Spam/ Dear, Friend})$ then we can answer the question asked

Email word	Spam	Email word	Spam
Dear	Yes	Friend	No
Friend	No	Friend	Yes
Dear	No	Dear	No
Dear	No	Lunch	No
Dear	No	Friend	No
Friend	No	Dear	No
Lunch	No	Dear	No
Friend	No	Dear	No
Lunch	No	Dear	No
Dear	Yes	Money	Yes
Money	Yes	Money	No
Money	Yes	Money	Yes

Solution:

Step 1- View or collect “raw” data.

Step 2 - Convert long data to a frequency table

	normal	spam	Row Total
word			
Dear	8	2	10
Friend	5	1	6
Lunch	3	0	3
Money	1	4	5
Column Total	17	7	24

Step 3 - Row and column sums to get probabilities

Email word	Spam	Email word	Spam
Dear	Yes	Friend	No
Friend	No	Friend	Yes
Dear	No	Dear	No
Dear	No	Lunch	No
Dear	No	Friend	No
Friend	No	Dear	No
Lunch	No	Dear	No
Friend	No	Dear	No
Lunch	No	Dear	No
Dear	Yes	Money	Yes
Money	Yes	Money	No
Money	Yes	Money	Yes

Solution:

Play	normal	spam	Row Total
word			
Dear	8	2	10
Friend	5	1	6
Lunch	3	0	3
Money	1	4	5
Column Total	17	7	24

As $P(N/D, F) > P(S/D, F)$, we can decide that Dear Friend is Normal message.

Step 4 - Apply probabilities from frequency table to Navie Bayes theorem

$$P(N/D, F) = \frac{P(D, F/N) \cdot P(N)}{P(D, F)} = \frac{P(D/N) \cdot P(F/N) \cdot P(N)}{P(D/N) \cdot P(F/N) \cdot P(N) + P(D/S) \cdot P(F/S) \cdot P(S)} = \frac{\left(\frac{8}{17}\right) \cdot \left(\frac{5}{17}\right) \cdot \left(\frac{17}{24}\right)}{\left(\frac{8}{17}\right) \cdot \left(\frac{5}{17}\right) \cdot \left(\frac{17}{24}\right) + \left(\frac{2}{7}\right) \cdot \left(\frac{1}{7}\right) \cdot \left(\frac{7}{24}\right)} = \frac{0.098}{0.11} = 0.891$$

$$P(S/D, F) = \frac{P(D, F/S) \cdot P(S)}{P(D, F)} = \frac{P(D/S) \cdot P(F/S) \cdot P(S)}{P(D/N) \cdot P(F/N) \cdot P(N) + P(D/S) \cdot P(F/S) \cdot P(S)} = \frac{\left(\frac{2}{7}\right) \cdot \left(\frac{1}{7}\right) \cdot \left(\frac{7}{24}\right)}{\left(\frac{8}{17}\right) \cdot \left(\frac{5}{17}\right) \cdot \left(\frac{17}{24}\right) + \left(\frac{2}{7}\right) \cdot \left(\frac{1}{7}\right) \cdot \left(\frac{7}{24}\right)} = \frac{0.012}{0.11} = 0.109$$

Question 4:

Suppose we got the new message contains the word '**Lunch Money Money Money Money**', Decide whether this new message is a normal or spam message?

Email word	Spam	Email word	Spam
Dear	Yes	Friend	No
Friend	No	Friend	Yes
Dear	No	Dear	No
Dear	No	Lunch	No
Dear	No	Friend	No
Friend	No	Dear	No
Lunch	No	Dear	No
Friend	No	Dear	No
Lunch	No	Dear	No
Dear	Yes	Money	Yes
Money	Yes	Money	No
Money	Yes	Money	Yes

Solution:

Play	normal	spam	Row Total
word			
Dear	8	2	10
Friend	5	1	6
Lunch	3	0	3
Money	1	4	5
Column Total	17	7	24

We can observe that we have to classify any message with Lunch as Normal message, no matter how many times we see the word Money and that's the problem.

Issue with Naïve Bayes classifier:
will not be able to classify as Normal or Spam!

Step 4 - Apply probabilities from frequency table

$$P(N). P(L/N). P(M/N)^4 = \left(\frac{17}{24}\right) \cdot \left(\frac{3}{17}\right) \cdot \left(\frac{1}{17}\right)^4 = 0.0000015$$

$$P(S). P(L/S). P(M/S)^4 = \left(\frac{7}{24}\right) \cdot \left(\frac{0}{7}\right) \cdot \left(\frac{4}{7}\right)^4 = 0$$

To work around this problem add 1 count to the frequency table to each word. **(Laplace smoothing)**

Solution:

Play	normal	spam	Row Total
word			
Dear	8+1	2+1	12
Friend	5+1	1+1	8
Lunch	3+1	0+1	5
Money	1+1	4+1	7
Column Total	21	11	32

As $P(S/L, M^4) > P(N/L, M^4)$, we can decide that **Lunch Money Money Money Money is Spam** message.

Step 4 - Apply probabilities from frequency table

$$P(N). P(L/N). P(M/N)^4 = \left(\frac{21}{32}\right) \cdot \left(\frac{4}{21}\right) \cdot \left(\frac{2}{21}\right)^4 = 0.00001$$

$$P(S). P(L/S). P(M/S)^4 = \left(\frac{11}{32}\right) \cdot \left(\frac{1}{11}\right) \cdot \left(\frac{5}{11}\right)^4 = 0.00133$$

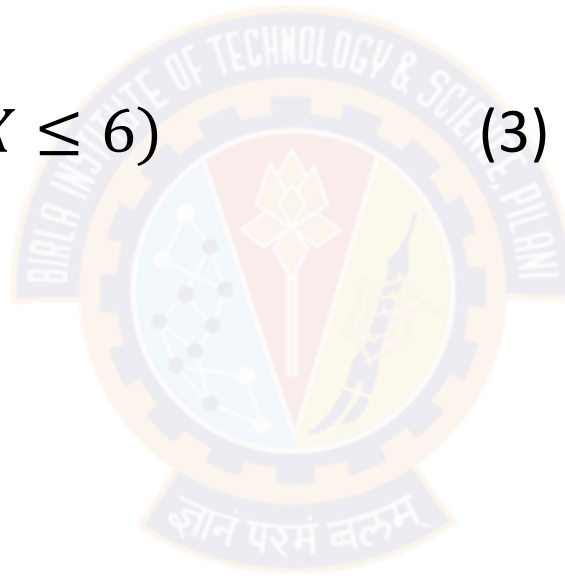
Question 5:

Let X be the random variable which represents the sum of numbers of points on throwing **two unbiased dice**. Construct the probability distribution for X and find the following probabilities:

(1) $P(X \geq 10)$

(2) $P(X \leq 6)$

(3) $P(7 < X < 12)$



Solution:

If the point shown in each die is equal to one, the minimum sum of numbers is equal to **2**.

If both dice show six, then the sum of numbers at the maximum is **12**.

Thus if two unbiased dice are thrown, then the sum **X** of the two numbers which turn up must be an integer between **2 and 12**.

For **X = 2**, there is only **one favourable sample point** (1, 1) and hence:

$$P(X = 2) = \frac{1}{36}$$

For **X = 3**, there are **two sample points**: (1, 2) and (2, 1).

Hence:

$$P(X = 3) = \frac{2}{36}$$

For **X = 4**, there are **three favourable sample points**: (1, 3), (2, 2), (3, 1).

Hence:

$$P(X = 4) = \frac{3}{36}$$

And so on.

Now the probability distribution in this case is given by the following table:

$X = x_i$	2	3	4	5	6	7	8	9	10	11	12
$P(X = x_i)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Clearly:

$$\sum p_i = p_1 + p_2 + \cdots + p_{11} = 1$$

Solution:

From the above table, we have:

1. $P(X \geq 10)$

$$P(X \geq 10) = \frac{3}{36} + \frac{2}{36} + \frac{1}{36} = \frac{6}{36} = \frac{1}{6}$$

2. $P(X \leq 6)$

$$\begin{aligned} P(X \leq 6) &= \frac{1}{36} + \frac{2}{36} + \frac{3}{36} + \frac{4}{36} + \frac{5}{36} + \frac{6}{36} \\ &= \frac{15}{36} = \frac{5}{12} \end{aligned}$$

3. $P(7 < X < 12)$

$$P(7 < X < 12) = \frac{5}{36} + \frac{4}{36} + \frac{3}{36} + \frac{2}{36} = \frac{14}{36} = \frac{7}{18}$$

Question 6:

Let $\mathbf{X}$ be a random variable with p.d.f.

$$f(x) = \begin{cases} ax & 0 \leq x \leq 1 \\ a & 1 \leq x \leq 2 \\ -ax + 3a & 2 \leq x < 3 \\ 0 & \text{otherwise} \end{cases}$$

(i) Determine the constant a

(ii) Compute $P(X \leq 1.5)$

Solution:

Since the total probability must be 1:

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

This becomes:

$$\int_0^1 f(x) dx + \int_1^2 f(x) dx + \int_2^3 f(x) dx = 1$$

Substitute each part:

$$\int_0^1 ax dx + \int_1^2 a dx + \int_2^3 (-ax + 3a) dx = 1$$

Compute each integral:

1.

$$\int_0^1 ax dx = a \left[\frac{x^2}{2} \right]_0^1 = \frac{a}{2}$$

2.

$$\int_1^2 a dx = a[x]_1^2 = a$$

Solution:

3.

$$\begin{aligned}\int_2^3 (-ax + 3a) dx &= a \left[\frac{-x^2}{2} + 3x \right]_2^3 \\&= a \left(-\frac{9}{2} + 9 - (-2 + 6) \right) \\&= a \left(-\frac{9}{2} + 9 + 2 - 6 \right) \\&= a \left(\frac{-9 + 18 - 12}{2} \right) = a\end{aligned}$$



So total probability:

$$\begin{aligned}\frac{a}{2} + a + a &= 1 \\ \frac{a}{2} + 2a &= 1 \\ \frac{5a}{2} &= 1 \\ a &= \frac{1}{2}\end{aligned}$$

Solution:

(ii) Compute $P(X \leq 1.5)$

$$P(X \leq 1.5) = \int_{-\infty}^{1.5} f(x) dx$$

This splits across two intervals:

$$= \int_0^1 f(x) dx + \int_1^{1.5} f(x) dx$$

Substitute $a = \frac{1}{2}$:

$$= \int_0^1 \frac{1}{2} x dx + \int_1^{1.5} \frac{1}{2} dx$$

Compute:

$$\int_0^1 \frac{1}{2} x dx = \frac{1}{2} \left[\frac{x^2}{2} \right]_0^1 = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

2.

$$\int_1^{1.5} \frac{1}{2} dx = \frac{1}{2} [x]_1^{1.5} = \frac{1}{2} (1.5 - 1) = \frac{1}{2} (0.5) = \frac{1}{4}$$

Solution:

Total:

$$P(X \leq 1.5) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

✓ Final Answers

(i) $a = \frac{1}{2}$

(ii) $P(X \leq 1.5) = \frac{1}{2}$



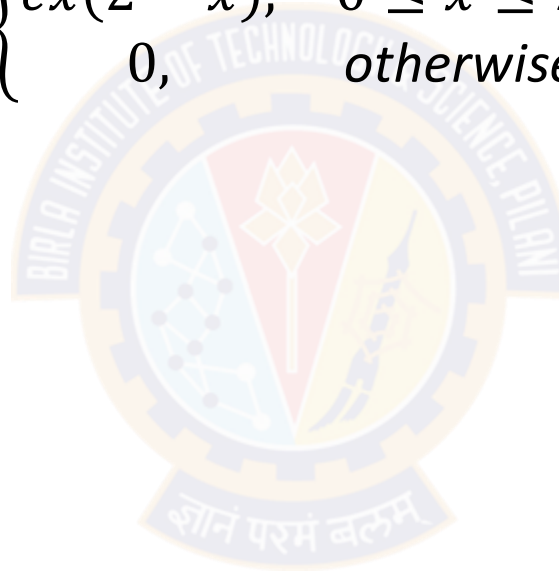
Question 7: Continuous Random Variable

- **Q7.** For the following probability density function:

$$f(x) = \begin{cases} cx(2-x), & 0 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

Find

- (i) the value of c
- (ii) mean
- (iii) variance.



Solution:

Since the total probability must be 1:

$$\int_0^2 cx(2-x) dx = 1$$

Compute the integral:

$$\begin{aligned} &= c \int_0^2 (2x - x^2) dx \\ &= c \left[x^2 - \frac{x^3}{3} \right]_0^2 \\ &= c \left(4 - \frac{8}{3} \right) \\ &= c \left(\frac{12 - 8}{3} \right) \\ &= c \cdot \frac{4}{3} \end{aligned}$$

So,

$$\begin{aligned} c \cdot \frac{4}{3} &= 1 \\ c &= \frac{3}{4} \end{aligned}$$

Thus, the probability density function becomes:

$$f(x) = \begin{cases} \frac{3}{4}x(2-x), & 0 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

Solution:

(ii) Mean of X : $E(X)$

$$\begin{aligned} E(X) &= \int_0^2 x f(x) dx \\ &= \int_0^2 x \cdot \frac{3}{4} x(2-x) dx \\ &= \frac{3}{4} \int_0^2 (2x^2 - x^3) dx \end{aligned}$$

Integrate:

$$\begin{aligned} &= \frac{3}{4} \left[\frac{2x^3}{3} - \frac{x^4}{4} \right]_0^2 \\ &= \frac{3}{4} \left(\frac{16}{3} - 4 \right) \end{aligned}$$

Thus:

$$\begin{aligned} &= \frac{3}{4} \left(\frac{16-12}{3} \right) \\ &= \frac{3}{4} \cdot \frac{4}{3} = 1 \end{aligned}$$

$$\boxed{E(X) = 1}$$

Solution:

(iii) *Variance of X*

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Compute $E(X^2)$:

$$\begin{aligned} E(X^2) &= \int_0^2 x^2 f(x) dx \\ &= \int_0^2 x^2 \cdot \frac{3}{4} x(2-x) dx \\ &= \frac{3}{4} \int_0^2 (2x^3 - x^4) dx \\ &= \frac{3}{4} \left[\frac{2x^4}{4} - \frac{x^5}{5} \right]_0^2 \\ &= \frac{3}{4} \left(8 - \frac{32}{5} \right) \end{aligned}$$

Convert to common denominator:

$$\begin{aligned} 8 &= \frac{40}{5} \\ 8 - \frac{32}{5} &= \frac{8}{5} \end{aligned}$$



Solution:

Thus:

$$E(X^2) = \frac{3}{4} \cdot \frac{8}{5} = \frac{24}{20} = \frac{6}{5}$$

Now use variance formula:

$$\begin{aligned} \text{Var}(X) &= E(X^2) - [E(X)]^2 \\ &= \frac{6}{5} - 1 \\ &= \frac{1}{5} \end{aligned}$$

Thus:

$$\boxed{\text{Var}(X) = \frac{1}{5}}$$



Question 8:

For a Binomial distribution with mean = 4 and variance = 2, find the probability of getting

(i) at least 2 successes

(ii) at most 2 successes



Solution:

The P.M.F. of a Binomial distribution is

$$P(X = x) = \binom{n}{x} p^x q^{n-x}$$

Given:

$$\begin{aligned} \text{Mean} &= np = 4 \\ \text{Variance} &= npq = 2 \end{aligned}$$

Step 1: Find p , q , and n

From variance:

Divide by mean:

$$npq = 2$$

$$\frac{npq}{np} = \frac{2}{4}$$

$$q = \frac{1}{2}$$

Hence:

$$p = 1 - q = 1 - \frac{1}{2} = \frac{1}{2}$$

Now use mean:

$$np = 4$$

$$n\left(\frac{1}{2}\right) = 4$$

$$n = 8$$

Solution:

Thus, the binomial distribution is:

$$P(X = x) = \binom{8}{x} \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{8-x}, x = 0, 1, \dots, 8.$$

(i) P (at least 2 successes)

$$\begin{aligned} P(X \geq 2) &= 1 - P(X < 2) \\ &= 1 - [P(X = 0) + P(X = 1)] \end{aligned}$$

Compute each term:

$$\begin{aligned} P(X = 0) &= \binom{8}{0} \left(\frac{1}{2}\right)^8 = \left(\frac{1}{2}\right)^8 \\ P(X = 1) &= \binom{8}{1} \left(\frac{1}{2}\right)^8 = 8 \left(\frac{1}{2}\right)^8 \end{aligned}$$

Thus:

$$\begin{aligned} P(X \geq 2) &= 1 - \left[\left(\frac{1}{2}\right)^8 + 8 \left(\frac{1}{2}\right)^8 \right] \\ &= 1 - 9 \left(\frac{1}{256}\right) \\ &= 1 - \frac{9}{256} \end{aligned}$$

$$\boxed{P(X \geq 2) = \frac{247}{256}}$$

Solution:

(ii) P(at most 2 successes)

$$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$$

We already have:

$$P(X = 0) = \left(\frac{1}{2}\right)^8$$

$$P(X = 1) = 8\left(\frac{1}{2}\right)^8$$

Now compute:

$$P(X = 2) = \binom{8}{2} \left(\frac{1}{2}\right)^8 = 28 \left(\frac{1}{2}\right)^8$$

Add:

$$P(X \leq 2) = \left(\frac{1}{2}\right)^8 + 8\left(\frac{1}{2}\right)^8 + 28\left(\frac{1}{2}\right)^8$$

$$= (1 + 8 + 28) \left(\frac{1}{256}\right)$$

$$= 37 \left(\frac{1}{256}\right)$$

$$\boxed{P(X \leq 2) = \frac{37}{256}}$$

Question 9

The probability of an individual suffering a bad reaction from an injection of a certain antibiotic is **0.001**.

Out of **2000 individuals**, find the probability that:

- (i) Exactly **3** suffer bad reaction
- (ii) More than **2** suffer bad reaction



Solution:

Let X denote the number of individuals suffering from bad reaction.

$$p = 0.001, n = 2000$$

Since n is large and the probability of reaction is small, we can assume X follows a **Poisson distribution** with

$$\lambda = np = (2000)(0.001) = 2.$$

Thus the Poisson probability function is:

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-2} 2^x}{x!}$$

(i) Probability that exactly 3 suffer

$$P(X = 3) = \frac{e^{-2} 2^3}{3!}$$

$$= \frac{e^{-2} \cdot 8}{6}$$

$$= 0.180447$$

$$\boxed{P(X = 3) \approx 0.180447}$$

Solution:

(ii) Probability that more than 2 suffer

$$\begin{aligned}P(X > 2) &= 1 - P(X \leq 2) \\&= 1 - [P(X = 0) + P(X = 1) + P(X = 2)] \\P(X > 2) &= 1 - [e^{-2} + 2e^{-2} + 2e^{-2}] \\&= 1 - 5e^{-2}\end{aligned}$$

Given:

$$\begin{aligned}e^{-2} &= 0.1353(\text{approx}) \\5e^{-2} &= 0.6767 \\P(X > 2) &= 1 - 0.6767 = 0.3233\end{aligned}$$

Final Answers

- (i) $P(X = 3) = 0.180447$
- (ii) $P(X > 2) = 0.3233$

Question 10:

From the following joint distribution of (X, Y) , find:

- (i) $P(X \leq 1)$
- (ii) $P(Y \leq 3)$
- (iii) $P(X \leq 1, Y \leq 3)$
- (iv) $P(X \leq 1 \mid Y \leq 3)$
- (v) $P(Y \leq 3 \mid X \leq 1)$
- (vi) $P(X + Y \leq 4)$

Joint Probability Table

$X \backslash Y$	1	2	3	4	5	6
0	0	0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{3}{32}$
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$
2	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{64}$	0	$\frac{2}{64}$

Solution:

Marginal probabilities (from the image):

$$P(X = 0) = \frac{8}{32}, P(X = 1) = \frac{20}{32}, P(X = 2) = \frac{4}{32}$$

$$P(Y = 1) = \frac{3}{32}, P(Y = 2) = \frac{3}{32}, P(Y = 3) = \frac{11}{64}, P(Y = 4) = \frac{13}{64}, P(Y = 5) = \frac{6}{32},$$

$$P(Y = 6) = \frac{16}{64}$$

(i) $P(X \leq 1)$

$$\begin{aligned} P(X \leq 1) &= P(X = 0) + P(X = 1) \\ &= \frac{8}{32} + \frac{20}{32} \\ &= \frac{28}{32} = \frac{7}{8} \end{aligned}$$

Solution:

(ii) $P(Y \leq 3)$

$$P(Y \leq 3) = P(Y = 1) + P(Y = 2) + P(Y = 3)$$

$$= \frac{3}{32} + \frac{3}{32} + \frac{11}{64}$$

$$= \frac{6}{32} + \frac{11}{64}$$

$$= \frac{12}{64} + \frac{11}{64}$$

$$= \frac{23}{64}$$

(iii) $P(X \leq 1, Y \leq 3)$

All entries with $X = 0, 1$ and $Y = 1, 2, 3$:

- $P(0, 1) = 0$

- $P(0, 2) = 0$

- $P(0, 3) = \frac{1}{32}$

- $P(1, 1) = \frac{1}{16}$

- $P(1, 2) = \frac{1}{16}$

- $P(1, 3) = \frac{1}{8}$

$$P(X \leq 1, Y \leq 3) = 0 + 0 + \frac{1}{32} + \frac{1}{16} + \frac{1}{16} + \frac{1}{8}$$

Convert to common denominator (32):

$$\begin{aligned} &= \frac{1}{32} + \frac{2}{32} + \frac{2}{32} + \frac{4}{32} \\ &= \frac{9}{32} \end{aligned}$$

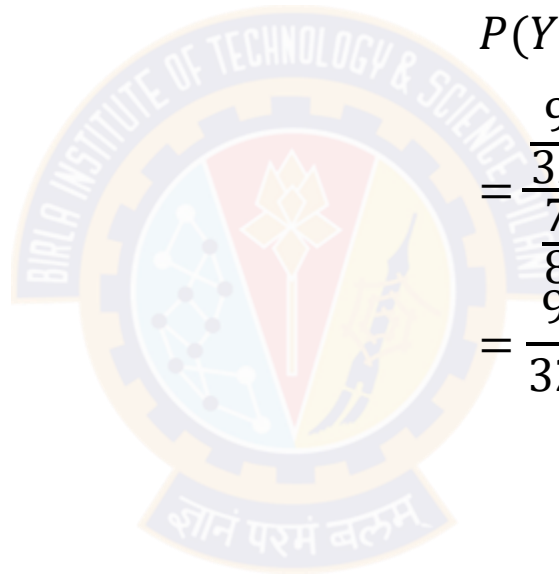
Solution:

(iv) $P(X \leq 1 \mid Y \leq 3)$

$$\begin{aligned} P(X \leq 1 \mid Y \leq 3) &= \frac{P(X \leq 1, Y \leq 3)}{P(Y \leq 3)} \\ &= \frac{\frac{9}{32}}{\frac{23}{64}} = \frac{9}{32} \cdot \frac{64}{23} \\ &= \frac{18}{23} \end{aligned}$$

(v) $P(Y \leq 3 \mid X \leq 1)$

$$\begin{aligned} P(Y \leq 3 \mid X \leq 1) &= \frac{P(X \leq 1, Y \leq 3)}{P(X \leq 1)} \\ &= \frac{\frac{9}{32}}{\frac{7}{8}} \\ &= \frac{9}{32} \cdot \frac{8}{7} = \frac{72}{224} = \frac{9}{28} \end{aligned}$$



Solution:

(vi) $P(X + Y \leq 4)$

We check all (x, y) pairs such that:

$$x + y \leq 4$$

Valid pairs from the table:

- $(0,1), (0,2), (0,3), (0,4)$
- $(1,1), (1,2), (1,3)$
- $(2,1), (2,2)$

Now sum their probabilities:

$$0 + 0 + \frac{1}{32} + \frac{2}{32} + \frac{1}{16} + \frac{1}{16} + \frac{1}{8} + \frac{1}{32} + \frac{1}{32}$$

Convert all to denominator 32:

$$= \frac{1}{32} + \frac{2}{32} + \frac{2}{32} + \frac{2}{32} + \frac{4}{32} + \frac{1}{32} + \frac{1}{32}$$

$$= \frac{13}{32}$$

$$\boxed{P(X + Y \leq 4) = \frac{13}{32}}$$

Final Answers Summary

Part	Answer
(i) $P(X \leq 1)$	$\frac{7}{8}$
(ii) $P(Y \leq 3)$	$\frac{23}{64}$
(iii) $P(X \leq 1, Y \leq 3)$	$\frac{9}{32}$
(iv) $P(X \leq 1 Y \leq 3)$	$\frac{18}{23}$
(v) $P(Y \leq 3 X \leq 1)$	$\frac{9}{28}$
(vi) $P(X + Y \leq 4)$	$\frac{13}{32}$



Question 11:

The joint p.d.f. of the random variable (X, Y) is given by

$$f(x, y) = \begin{cases} \frac{x(1 + 3y^2)}{4}, & 0 < x < 2, 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find:

- (i) Marginal density functions of X and Y
- (ii) Conditional density of X given Y
- (iii) $P\left(\frac{1}{4} < X < \frac{1}{2} \mid Y = \frac{1}{3}\right)$



Solution:

(i) Marginal Density Functions

Marginal density of X :

$$\begin{aligned}f_X(x) &= \int_0^1 f(x, y) dy \\&= \int_0^1 \frac{x(1 + 3y^2)}{4} dy \\&= \frac{x}{4} \int_0^1 (1 + 3y^2) dy \\&= \frac{x}{4} [y + y^3]_0^1 \\&= \frac{x}{4} (1 + 1) = \frac{x}{2}\end{aligned}$$

Thus,

$$f_X(x) = \frac{x}{2}, 0 < x < 2$$

Solution:

Marginal density of Y:

$$\begin{aligned} f_Y(y) &= \int_0^2 f(x, y) dx \\ &= \int_0^2 \frac{x(1 + 3y^2)}{4} dx \\ &= \frac{1 + 3y^2}{4} \int_0^2 x dx \\ &= \frac{1 + 3y^2}{4} \left[\frac{x^2}{2} \right]_0^2 \\ &= \frac{1 + 3y^2}{4} \cdot \frac{4}{2} = \frac{1 + 3y^2}{2} \end{aligned}$$

Thus,

$$f_Y(y) = \frac{1 + 3y^2}{2}, 0 < y < 1$$

Solution:

(ii) Conditional Density of X given $Y = y$

$$f_{X|Y}(x | y) = \frac{f(x, y)}{f_Y(y)}$$

$$= \frac{\frac{x(1 + 3y^2)}{4}}{\frac{1 + 3y^2}{2}}$$

$$= \frac{x}{4} \cdot \frac{2}{1} = \frac{x}{2}$$

Thus,

$$f_{X|Y}(x | y) = \frac{x}{2}, 0 < x < 2$$



Solution:

(iii) Compute

$$P\left(\frac{1}{4} < X < \frac{1}{2} \mid Y = \frac{1}{3}\right)$$

Since conditional density does not depend on y :

$$\begin{aligned} P &= \int_{1/4}^{1/2} \frac{x}{2} dx \\ &= \frac{1}{2} \int_{1/4}^{1/2} x dx \\ &= \frac{1}{2} \left[\frac{x^2}{2} \right]_{1/4}^{1/2} \\ &= \frac{1}{2} \left(\frac{(1/2)^2}{2} - \frac{(1/4)^2}{2} \right) \\ &= \frac{1}{2} \left(\frac{1}{8} - \frac{1}{32} \right) \\ &= \frac{1}{2} \cdot \frac{3}{32} = \frac{3}{64} \end{aligned}$$

$$\boxed{P\left(\frac{1}{4} < X < \frac{1}{2} \mid Y = \frac{1}{3}\right) = \frac{3}{64}}$$

Question 12:

For a normally distributed variable with mean 1 and standard deviation 3, find the probabilities that

(i) $3.43 \leq x \leq 6.19$

(ii) $-1.43 \leq x \leq 6.19$



Solution:

(i) When $x = 3.43$,

$$z = \frac{x-\mu}{\sigma} = \frac{3.43-1}{3} = \frac{2.43}{3} = 0.81 = z_1 \text{ (say)}$$

When $x = 6.19$,

$$z = \frac{x-\mu}{\sigma} = \frac{6.19-1}{3} = \frac{5.19}{3} = 1.73 = z_2 \text{ (say)}$$

Therefore,

$$P(3.43 \leq x \leq 6.19) = P(0.81 \leq z \leq 1.73)$$

$$= [F(z_2) - F(z_1)]$$

$$= [F(1.73) - F(0.81)]$$

$$= 0.4582 - 0.2910 \text{ (from normal tables)} = 0.1672$$

Solution:

(ii) When $x = -1.43$,

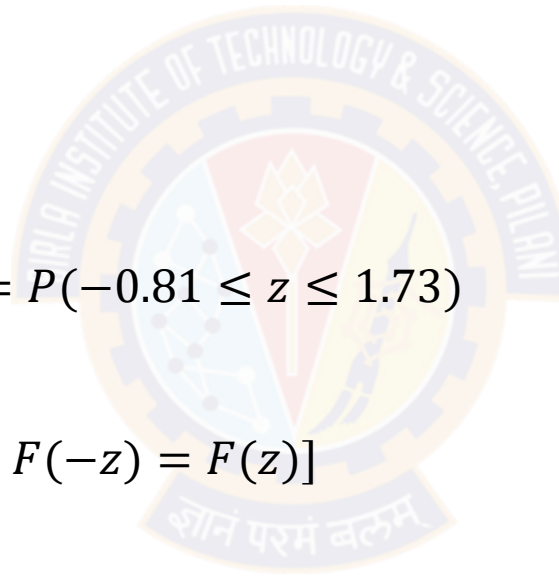
$$z = \frac{x-\mu}{\sigma} = \frac{-1.43-1}{3} = -0.81 = z_1 \text{ (say)}$$

When $x = 6.19$,

$$z = \frac{x-\mu}{\sigma} = \frac{6.19-1}{3} = 1.73 = z_2 \text{ (say)}$$

Therefore,

$$\begin{aligned} P(-1.43 \leq x \leq 6.19) &= P(-0.81 \leq z \leq 1.73) \\ &= F(z_2) + F(z_1) \\ &= F(1.73) + F(-0.81) \\ &= F(1.73) + F(0.81) [\because F(-z) = F(z)] \\ &= 0.4582 + 0.2910 \\ &= 0.7492 \end{aligned}$$



Question 13

If X is normally distributed with mean 1 and standard deviation 0.6, obtain

- (i) $P(X > 0)$ and
- (ii) $P(-1.8 \leq X \leq 2.0)$



Solution:

We have, mean $\mu = 1$ and S.D. $\sigma = 0.6$

The standard normal variable $N(0,1)$ is

$$z = \frac{X - \mu}{\sigma}$$

(i) When $X > 0$,

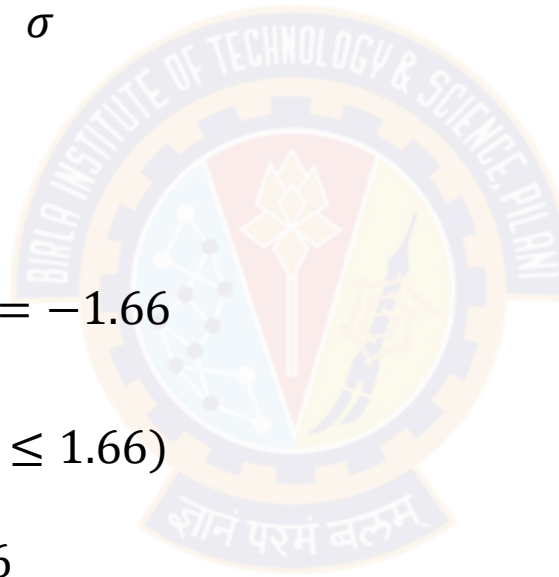
$$z = \frac{X - \mu}{\sigma} = \frac{X - 1}{0.6} = \frac{-1}{0.6} = -1.66$$

$$P(X > 0)$$

$$= P(z > 1.66) = 1 - P(z \leq 1.66)$$

$$= 1 - F(1.66)$$

$$= 1 - 0.95154 = 0.04846$$



Solution:

(ii) When $X = -1.8$,

$$Z = \frac{-1.8 - 1}{0.6} = \frac{-2.8}{0.6} = \frac{-2.8}{6} = -4.66$$

When $X = 2.0$,

$$Z = \frac{2 - 1}{0.6} = \frac{1}{0.6} = 1.66$$

Therefore,

$$\begin{aligned} P(-1.8 \leq X \leq 2.0) &= P(-4.66 < z < 1.66) \\ &= F(1.66) - F(-4.66) \\ &= 0.95154 - 0 = 0.95154 \end{aligned}$$



THANK YOU